

Triggers & Stored Procedures

Rayanki Himavarshini (2520030141)

Question : Develop a Bank Database using Customer, Account and Bank_Transaction tables. Implement stored procedures for customer and account operations, deposits, withdrawals, money transfer and balance retrieval. Implement BEFORE and AFTER triggers for balance validation, transaction validation, transaction auditing, automatic balance updates and insufficient-balance prevention.

Step 1 : Create Database and Table

The screenshot shows the MySQL Workbench interface. The SQL Editor window displays the following SQL script:

```
1  -- =====
2  -- BANK TRIGGER PROJECT
3  -- =====
4
5  -- Step 1: Create Database
6  DROP DATABASE IF EXISTS Triggers_DB;
7  CREATE DATABASE Triggers_DB;
8  USE Triggers_DB;
9
```

The Output window shows the execution results of the SQL script:

#	Time	Action	Message	Duration / Fetch
1	12:25:50	DROP PROCEDURE IF EXISTS HighBalanceAccounts	Error Code: 1046. No database selected Select the default DB to be used b...	0.000 sec
2	12:25:52	CALL HighBalanceAccounts(100000)	Error Code: 1046. No database selected Select the default DB to be used b...	0.000 sec
3	12:27:13	DROP DATABASE IF EXISTS Triggers_DB	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'triggers_db'; data...	0.016 sec
4	12:27:13	CREATE DATABASE Triggers_DB	1 row(s) affected	0.015 sec
5	12:27:13	USE Triggers_DB	0 row(s) affected	0.000 sec

```

6 -- CREATE DATABASE Triggers_DB;
7 • USE Triggers_DB;
8 -- =====
9 -- Step 2: Create Customer Table
10 -- =====
11 • CREATE TABLE Customer (
12     Customer_ID INT PRIMARY KEY,
13     Customer_Name VARCHAR(100) NOT NULL,
14     Phone VARCHAR(15),
15     Email VARCHAR(100),
16     City VARCHAR(50)
17 );
18

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
❌ 1	12:25:50	DROP PROCEDURE IF EXISTS HighBalanceAccounts	Error Code: 1046. No database selected Select the default DB to be used b...	0.000 sec
❌ 2	12:25:52	CALL HighBalanceAccounts(100000)	Error Code: 1046. No database selected Select the default DB to be used b...	0.000 sec
⚠️ 3	12:27:13	DROP DATABASE IF EXISTS Triggers_DB	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'triggers_db'; data...	0.016 sec
✅ 4	12:27:13	CREATE DATABASE Triggers_DB	1 row(s) affected	0.015 sec
✅ 5	12:27:13	USE Triggers_DB	0 row(s) affected	0.000 sec
✅ 6	12:34:28	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Custo...	0 row(s) affected	0.047 sec

```

18 -- =====
19 -- Step 3: Create Account Table
20 -- =====
21 • CREATE TABLE Account (
22     Account_No INT PRIMARY KEY,
23     Customer_ID INT,
24     Account_Type VARCHAR(20),
25     Balance DECIMAL(12,2) DEFAULT 0,
26     Branch VARCHAR(50),
27
28     FOREIGN KEY (Customer_ID)
29     REFERENCES Customer(Customer_ID)
30 );

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
❌ 2	12:25:52	CALL HighBalanceAccounts(100000)	Error Code: 1046. No database selected Select the default DB to be used...	0.000 sec
⚠️ 3	12:27:13	DROP DATABASE IF EXISTS Triggers_DB	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'triggers_db'; da...	0.016 sec
✅ 4	12:27:13	CREATE DATABASE Triggers_DB	1 row(s) affected	0.015 sec
✅ 5	12:27:13	USE Triggers_DB	0 row(s) affected	0.000 sec
✅ 6	12:34:28	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Custo...	0 row(s) affected	0.047 sec
✅ 7	12:35:24	CREATE TABLE Account (Account_No INT PRIMARY KEY, Custo...	0 row(s) affected	0.047 sec

31 -- Step 4: Create Bank Transaction Table
32 -- =====
33 • CREATE TABLE Bank_Transaction (
34 Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
35 Account_No INT,
36 Transaction_Type VARCHAR(20),
37 Amount DECIMAL(12,2),
38 Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
39
40 FOREIGN KEY (Account_No)
41 REFERENCES Account(Account_No)
42);
43

Automatic context help is disabled. Use the toolbar manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	12:27:13	DROP DATABASE IF EXISTS Triggers_DB	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'triggers_db'; da...	0.016 sec
4	12:27:13	CREATE DATABASE Triggers_DB	1 row(s) affected	0.015 sec
5	12:27:13	USE Triggers_DB	0 row(s) affected	0.000 sec
6	12:34:28	CREATE TABLE Customer(Customer_ID INT PRIMARY KEY, Cust...	0 row(s) affected	0.047 sec
7	12:35:24	CREATE TABLE Account(Account_No INT PRIMARY KEY, Custo...	0 row(s) affected	0.047 sec
8	12:36:17	CREATE TABLE Bank_Transaction(Transaction_ID INT PRIMARY K...	0 row(s) affected	0.032 sec

74 REFERENCES Account(Account_No)
42);
43 -- =====
44 -- Step 5: Insert Customer Data
45 -- =====
46 • INSERT INTO Customer VALUES
47 (101,'Hima','2520030141','hima@gmail.com','Hyderabad'),
48 (102,'Ravi','2520030126','ravi@gmail.com','Vijayawada'),
49 (103,'Santhi','9876543212','santhin@gmail.com','Bangalore'),
50 (104,'Siri','9876543213','siri@gmail.com','Chennai'),
51 (105,'Varshini','9876543214','varshini@gmail.com','Hyderabad');

manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	12:27:13	CREATE DATABASE Triggers_DB	1 row(s) affected	0.015 sec
5	12:27:13	USE Triggers_DB	0 row(s) affected	0.000 sec
6	12:34:28	CREATE TABLE Customer(Customer_ID INT PRIMARY KEY, Cust...	0 row(s) affected	0.047 sec
7	12:35:24	CREATE TABLE Account(Account_No INT PRIMARY KEY, Custo...	0 row(s) affected	0.047 sec
8	12:36:17	CREATE TABLE Bank_Transaction(Transaction_ID INT PRIMARY K...	0 row(s) affected	0.032 sec
9	12:38:14	INSERT INTO Customer VALUES (101,'Hima','2520030141','hima@gmail...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

52

-- =====

53

-- Step 6: Insert Account Data

54

-- =====

55

• **INSERT INTO Account VALUES**

56

(10001,101,'Savings',50000,'Hyderabad'),

57

(10002,102,'Savings',75000,'Vijayawada'),

58

(10003,103,'Current',120000,'Bangalore'),

59

(10004,104,'Savings',45000,'Chennai'),

60

(10005,105,'Current',90000,'Hyderabad');

toggle automatic help

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	12:27:13	USE Triggers_DB	0 row(s) affected	0.000 sec
6	12:34:28	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Cust...	0 row(s) affected	0.047 sec
7	12:35:24	CREATE TABLE Account (Account_No INT PRIMARY KEY, Custo...	0 row(s) affected	0.047 sec
8	12:36:17	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY K...	0 row(s) affected	0.032 sec
9	12:38:14	INSERT INTO Customer VALUES (101,'Hima','2520030141','hima@gmail...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
10	12:39:33	INSERT INTO Account VALUES (10001,101,'Savings',50000,'Hyderaba...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec

SQL File 1

Procedures(Question_4)

Limit to 1000 rows

SQL Additions

59

(10004,104,'Savings',45000,'Chennai'),

60

(10005,105,'Current',90000,'Hyderabad');

61

-- =====

62

-- Step 7: Create Transaction Audit Table

63

-- =====

64

• **CREATE TABLE Transaction_Audit (**

65

Audit_ID INT PRIMARY KEY AUTO_INCREMENT,

66

Transaction_ID INT,

67

Account_No INT,

68

Transaction_Type VARCHAR(20),

69

Amount DECIMAL(12,2),

70

Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP

71

);

Automatic context help disabled. Use the toolbar manually get help for th current caret position or toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	12:34:28	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Cust...	0 row(s) affected	0.047 sec
7	12:35:24	CREATE TABLE Account (Account_No INT PRIMARY KEY, Custo...	0 row(s) affected	0.047 sec
8	12:36:17	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY K...	0 row(s) affected	0.032 sec
9	12:38:14	INSERT INTO Customer VALUES (101,'Hima','2520030141','hima@gmail...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
10	12:39:33	INSERT INTO Account VALUES (10001,101,'Savings',50000,'Hyderaba...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	12:40:25	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AU...	0 row(s) affected	0.032 sec

SQL File 1 Procedures[Question_4]

Limit to 1000 rows

```

74  -- =====
75  DELIMITER //
76  CREATE TRIGGER CheckBalance
77  BEFORE UPDATE ON Account
78  FOR EACH ROW
79  BEGIN
80      IF NEW.Balance < 0 THEN
81          SIGNAL SQLSTATE '45000'
82          SET MESSAGE_TEXT =
83              'Transaction failed: Insufficient balance';
84      END IF;
85  END //
86  DELIMITER ;

```

Context Help

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message
7	12:35:24	CREATE TABLE Account(Account_No INT PRIMARY KEY, Custo...	0 row(s) affected
8	12:36:17	CREATE TABLE Bank_Transaction(Transaction_ID INT PRIMARY K...	0 row(s) affected
9	12:38:14	INSERT INTO Customer VALUES (101,'Hima','2520030141','hima@gmail...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
10	12:39:33	INSERT INTO Account VALUES (10001,101,'Savings',50000,'Hyderaba...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
11	12:40:25	CREATE TABLE Transaction_Audit(Audit_ID INT PRIMARY KEY AU...	0 row(s) affected
12	12:41:41	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR...	0 row(s) affected

SQL File 1 Procedures[Question_4]

Limit to 1000 rows

```

89  -- =====
90  DELIMITER //
91  CREATE TRIGGER CheckTransactionAmount
92  BEFORE INSERT ON Bank_Transaction
93  FOR EACH ROW
94  BEGIN
95      IF NEW.Amount <= 0 THEN
96          SIGNAL SQLSTATE '45000'
97          SET MESSAGE_TEXT =
98              'Transaction amount must be greater than zero';
99      END IF;
100  END //
101  DELIMITER ;

```

Context Help Snippets

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	12:36:17	CREATE TABLE Bank_Transaction(Transaction_ID INT PRIMARY K...	0 row(s) affected	0.032 sec
9	12:38:14	INSERT INTO Customer VALUES (101,'Hima','2520030141','hima@gmail...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
10	12:39:33	INSERT INTO Account VALUES (10001,101,'Savings',50000,'Hyderaba...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	12:40:25	CREATE TABLE Transaction_Audit(Audit_ID INT PRIMARY KEY AU...	0 row(s) affected	0.032 sec
12	12:41:41	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR...	0 row(s) affected	0.015 sec
13	12:43:08	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Ba...	0 row(s) affected	0.016 sec

SQL File 1 Procedures(Question_4) x

Limit to 1000 rows

```

110 DECLARE CurrentBalance DECIMAL(12,2);
111 SELECT Balance
112 INTO CurrentBalance
113 FROM Account
114 WHERE Account_No = NEW.Account_No;
115 IF NEW.Transaction_Type = 'WITHDRAW'
116     AND NEW.Amount > CurrentBalance THEN
117     SIGNAL SQLSTATE '45000'
118     SET MESSAGE_TEXT =
119     'Withdrawal failed: Insufficient balance';
120 END IF;
121 END //
122 DELIMITER ;

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	12:38:14	INSERT INTO Customer VALUES (101,'Hima','2520030141','hima@gmail...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
10	12:39:33	INSERT INTO Account VALUES (10001,101,'Savings',50000,Hyderaba...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	12:40:25	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AU...	0 row(s) affected	0.032 sec
12	12:41:41	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR...	0 row(s) affected	0.015 sec
13	12:43:08	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Ba...	0 row(s) affected	0.016 sec
14	12:44:28	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON...	0 row(s) affected	0.015 sec

Limit to 1000 rows

```

134 Account_No,
135 Transaction_Type,
136 Amount
137 )
138 VALUES
139 (
140     NEW.Transaction_ID,
141     NEW.Account_No,
142     NEW.Transaction_Type,
143     NEW.Amount
144 );
145 END //
146 DELIMITER ;

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	12:39:33	INSERT INTO Account VALUES (10001,101,'Savings',50000,Hyderaba...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	12:40:25	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AU...	0 row(s) affected	0.032 sec
12	12:41:41	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR...	0 row(s) affected	0.015 sec
13	12:43:08	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Ba...	0 row(s) affected	0.016 sec
14	12:44:28	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON...	0 row(s) affected	0.015 sec
15	12:45:33	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Tran...	0 row(s) affected	0.016 sec